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And how to use AI to enhance our workflow

Api testing
Automation basics.

Agenda

01 What is an API

02 Cohn pyramid and why API testing

03 How can we test api

04 Why automate api

05 Api framework layer proposal an explanation

06 Tools

07 How to automate api (workshop)

08 Using IA to generate tests

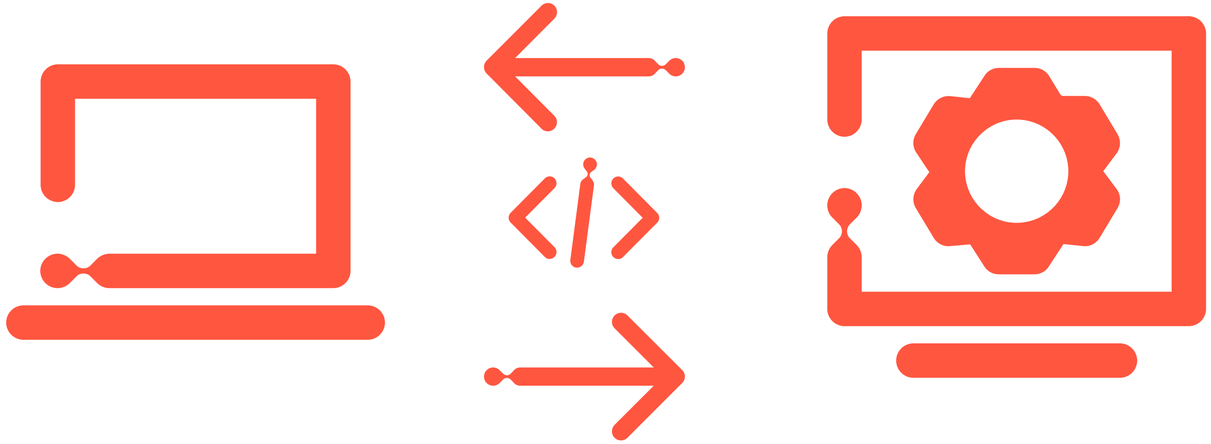
And why it matters

What is an API?

Chapter

01 Aplication Programing Interface

An application programing Interface is a set of rules , protocols and definitions that allows for one programing application to communicate with another



Chapter

01 Most typical API by architecture

Rest API: uses an http protocol to access and manipulate resources by leveraging the http methods, in it a Rest sever receives a request and provides a response.

Know your audience

Parts of a REST API request

Endpoint

Its a way to identify a request for which and action is to be taken. It is done by an URL

Method

It's a verb that describes the desired action that is to be taken on a specific resource

Body

Specifies additional information that needs to be used upon the resource or method.

Headers

Specifies additional information that needs to be used upon the communicating. authentication information

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Know your audience

Parts of a REST API response

Status code

It's a code that explains in lesser detail the outcome of the desire operation.

Method

It's a verb that describes the desired action that is to be taken on a specific resource

Body

Detailed information about the result of the process as well as retrieves the information required if necessary

Headers

Gicves ditional information about the communication used for the response

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Manually without code

How:
To test and API.

Chapter

01 How To test and api

For testing an api we use a tool that allow is to create a request. It can be a curl command or an application like Bruno or postman

API Testing Heuristic

POISED

Parameters

Are the different parameters commonly used as part for the request

Output

Are the different response elements, status code, body message, headers,etc.

Interoperability

Is the ability to provide other services what they need how the need it

Security

Is the ability of the service to withstand, attempts to bypass desired access or other type of vulnerabilities

Errors

Is the capacity to properly behave on unexpected situations

Data

Is the handling of the data, the data must be accurate fetched and stored

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It makes life easier for projects and you

Why Automate API tests?.

Know your audience

Advantages of automation

Reusability
The tests can be executed more often

Time saving
The execution is way faster, allowing more tests executed in less time

Accuracy
Automation is less prone to errors than human execution

Cost efficiency
If well made the test automation will save resources by allowing faster and more reliable bug detention

CI/CD integration
It allows for integration with CI/CD approaches, which in time means faster time to market

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Automation Layer proposal.

API Testing Heuristic

Test framework layers

Test

In this layers we manage and create the different tests

Service

This layer is responsible for creating and managing the request of the different services

API Client

This Layer is responsible for managing the request logic

Data

This layer maintains and create the data needed for the tests

Utils

This Layer manages different utilities that may come in handy when working on tests.

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Know your audience

Tools

Know your audience

Tools

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Know your audience

Tools

Playwright

Playwright Test is a modern End-to-End testing framework that has gained a lot of traction in the past few years. It is developed by Microsoft and offers fast isolated tests that are compatible with modern technologies

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WorkShop

How :
To automate api?.

Work in Progress!!

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Know your audience

Homework

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Using IA :
For test generation.

For our use of AI tools

Important concepts

CLI

Command line interface, are tools that allow is to perform different actions by using commands on a command prompt

Skills

Are files used to provide an AI agent with a knowledge reference of how to perform an specific tasks, by doing so the agent doesn't need to keep a huge amount on information Inside the context window

Rules

Are guidelines for the AI behavior, allowing us to place restrictions on how the agent will behave e.g. Don't deploy to prod

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Back to work!!

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Yay

Homework

Create a framework
for this task you will need to create a framework and add tests to them.

1. **Search for a public API that can be used for testing**
   search for one or more api that are public and can be used to tests
2. **Define tests framework layers**
   choose a set of layers that you plan to include and why this must be a word document attached to the framework.
3. **Incorporate tests**
   following poised create tests cases for request that atleat have 1 post, 1 get and 1 put method. In total you must bring a minimum of 9 tests
4. **Upload to GitHub**
   your framework must be uploaded on a public github repo and the link should se sent as part of the homework.
5. **Document your usage of IA**
   all the help you took form IA tools must be documented and explained

All written artifacts that are not code must be uploaded on a folder for the homework. The Link to the repository must be included at the star of each of them.

The Homework will be evaluated on completion and quality. Each of the numerals above will be 20% of the total score.

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Thank you!!!

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